

Representation of Data

Answer Booklet



Worked solutions for the 20-question Chapter 1 practice set.

SECTION A - STEM-AND-LEAF DIAGRAMS

A1 Coin bags and maximum value

Leaves	Stem	
0 2 6 8 9	15	
0 2 3 5	16	
0 2 5	17	

Key: 15 | 0 represents 150 coins

(b) To maximise the total value, use the three lowest bag counts for the \$0.10, \$0.25 and \$0.50 bags, and make the remaining nine bags \$1 bags. Value = $(150 \times 0.10) + (152 \times 0.25) + (156 \times 0.50) + (1484 \times 1.00) = \1615 .

A2 Ferry passengers

(a) Ferry Z total = 649 and ferry A total = 561, so $649 - 561 = 88$ more passengers travelled on ferry Z.

(b) $(561 \times 12.50) - 649x = 3.30$, giving $x = \$10.80$.

(c) The common passenger totals visible in the two diagrams are 30, 43 and 45. They may or may not have occurred on the same days. Least possible = 0 days; greatest possible = 3 days.

A3 Comparing two batsmen

Batsman P	Stem	Batsman Q
	2	0 1
9 8 7 7 6	3	1 6
8 7 4 1 1	4	2 5 8
9 9 7 3 2	5	1 2 6 7
	6	4 8
	7	1 7

Key: 6 | 3 | 1 represents 36 runs for P and 31 runs for Q

(b)(i) Totals: P = 688, Q = 739. Therefore Q performed better by total runs.

(b)(ii) P's scores run from 36 to 59 (range 23), while Q's run from 20 to 77 (range 57). P was more consistent because the scores are less spread out.

A4 Bird nests: rows of width 5

Wrens (10)	Stem	Dunnocks (10)
3	1	
9 8 7	1	7 9
4 3 3 2 1 0	2	2 2 3 4
	2	5 7 8
	3	0

Key: 8 | 1 | 9 represents 18 eggs for a wren and 19 eggs for a dunnock

(b) Total dunnock eggs = 237. Estimated survivors = $0.92 \times 237 = 218$ chicks (nearest whole chick).

(c) Total wren eggs = 200. If 14 did not survive, 186 survived. Survival rate = $(186 / 200) \times 100\% = 93\%$.

A5 Top examination scores

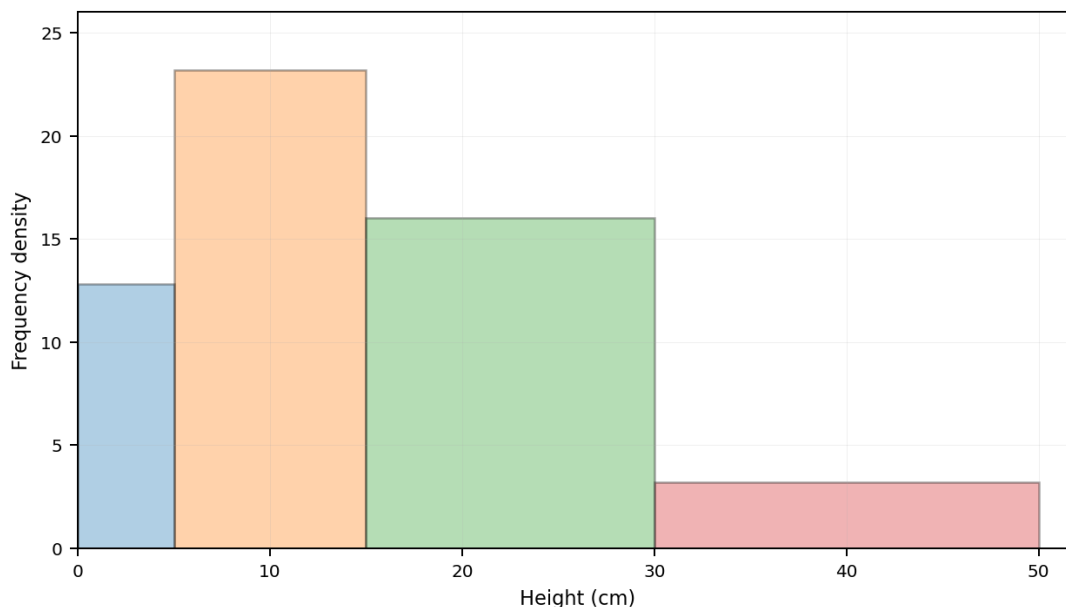
The middle student is the 13th when the 25 scores are arranged in order. This is the girl who scored 92%.

By placing the girl who scored 93% to the left of the two boys who scored 93%, a block of seven boys can be formed with no girls between them. The greatest possible number of boys who are not standing next to a girl is 5.

SECTION B - HISTOGRAMS AND FREQUENCY

B1 Heights of saplings

(a) Any suitable u from 35 to 50 is acceptable. Using $u = 50$ gives the histogram shown.

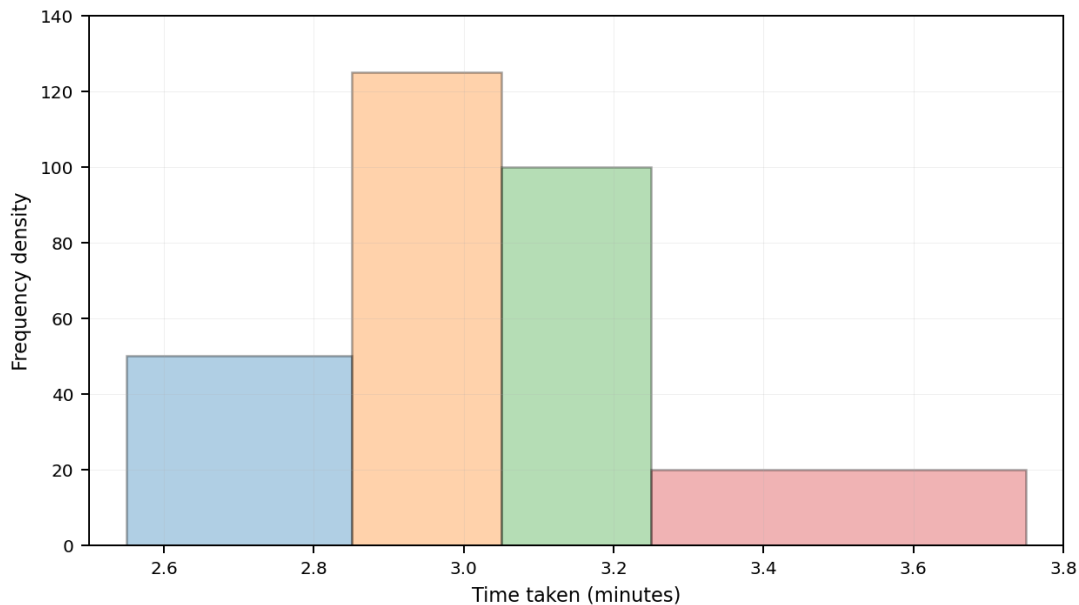


(c)(i) Less than 25 cm: $64 + 232 + [(25 - 15)/(30 - 15)] \times 240 = 456$ saplings.

(c)(ii) Between 7.5 and 19.5 cm: $[(15 - 7.5)/(15 - 5)] \times 232 + [(19.5 - 15)/(30 - 15)] \times 240 = 174 + 72 = 246$ saplings.

B2 Typing times

(a) The first class actually runs from 2.55 to 2.85 minutes because values are correct to the nearest 0.1 min.
Width = $2.85 - 2.55 = 0.30$ min.



(c) The fastest 10 trainees are $10/15 = 2/3$ of the way through the first class: $2.55 + (2/3)(0.30) = 2.75$ min = 165 s.

(d)(i) For $b > 3.15$: 10 of the 15 trainees lie in the third class, so 5 must lie in the fourth class. Half of the 3.25-3.75 class gives $b = 3.50$.

(d)(ii) For $b < 3.15$: $b = 3.05 - (5/25)(3.05 - 2.85) = 3.01$ minutes.

B3 Equal-height histogram columns

(a) Equal column height means equal frequency density. The class widths are 3, 7, 20 and 12 kg. Frequency density = $371/7 = 1060/20 = 53$. Hence $a = 53 \times 3 = 159$ and $b = 53 \times 12 = 636$. Total = $159 + 371 + 1060 + 636 = 2226$.

(b) 50% of 2226 is 1113, so use the 1114th mass. It lies in the 13-32 kg class. Position within that class = $1114 - 159 - 371 = 584$. Estimate = $12.5 + (584/1060)(32.5 - 12.5) = 23.5$ kg.

B4 Frequency-density and width ratios

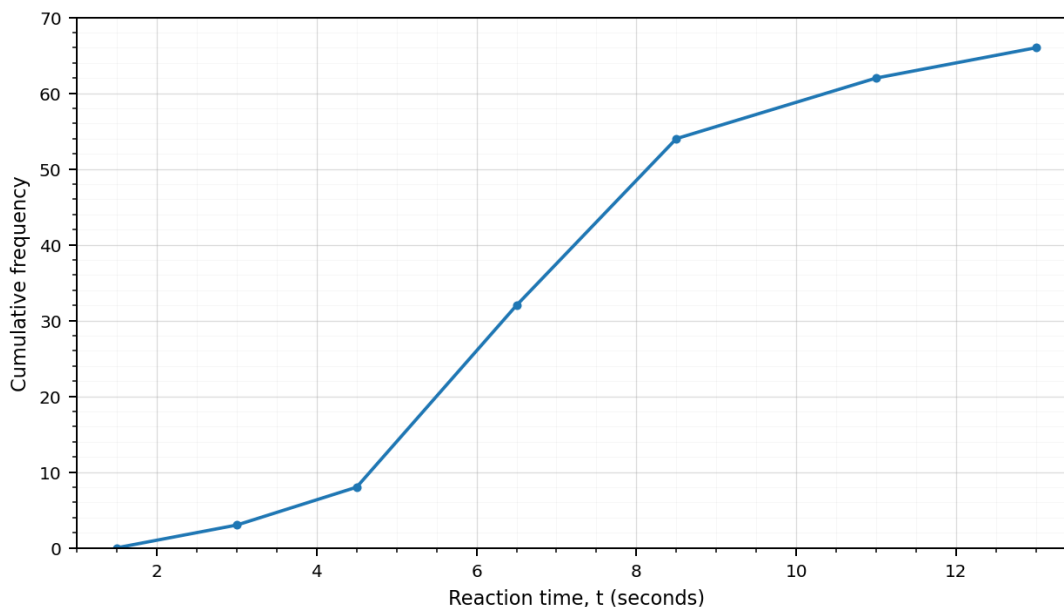
Class width is proportional to frequency / frequency density. Width ratio = $10/4 : 15/3 : 24/2 : 8/1 = 2.5 : 5 : 12 : 8$. Scale so the narrowest width is 3 cm: $3 : 6 : 14.4 : 9.6$. Total width = 33 cm.

B5 Missing class limits

The class widths are $51 - p$, 20, 10 and $q - 80$. Since the second class has frequency density $240/20 = 12$ and this corresponds to ratio 8, the four densities are 7.5, 12, 19.5 and 10.5. Then $165/(51 - p) = 7.5$ gives $p = 29$, and $147/(q - 80) = 10.5$ gives $q = 94$.

SECTION C - CUMULATIVE FREQUENCY

C1 Reaction times



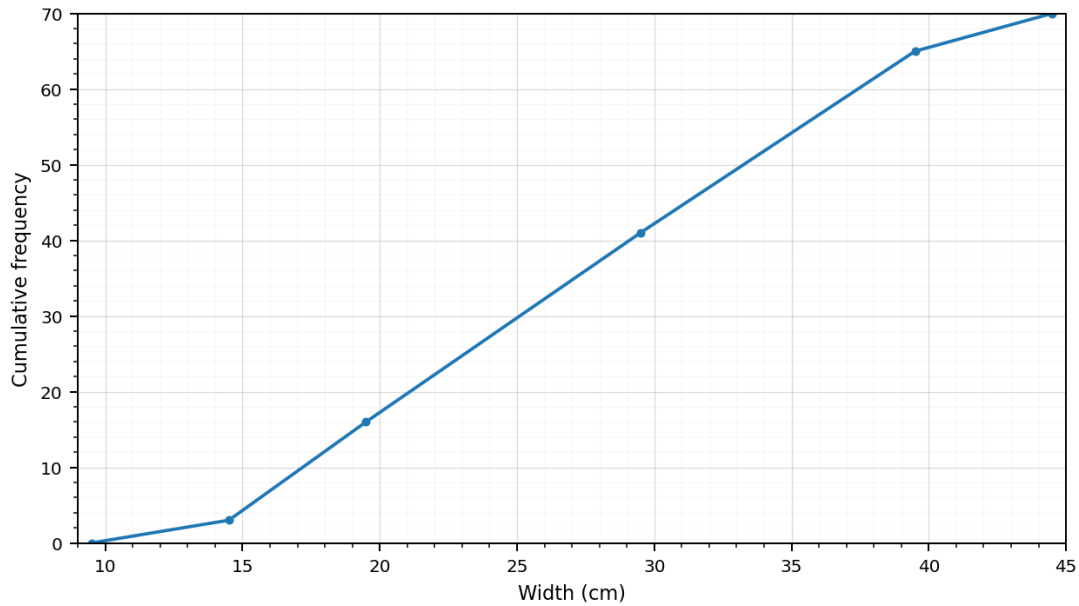
(b)(i) From the polygon, cf at 7.5 s is about 43 and cf at 5.5 s is about 20. Estimate = $43 - 20 = 23$ participants.

(b)(ii) The slowest 20 are above the $66 - 20 = 46$ th value. The graph gives the 46th value at about 7.8 s.

C2 Widths of books

(a) Upper boundary of the second class = $19 + 0.5 = 19.5$ cm.

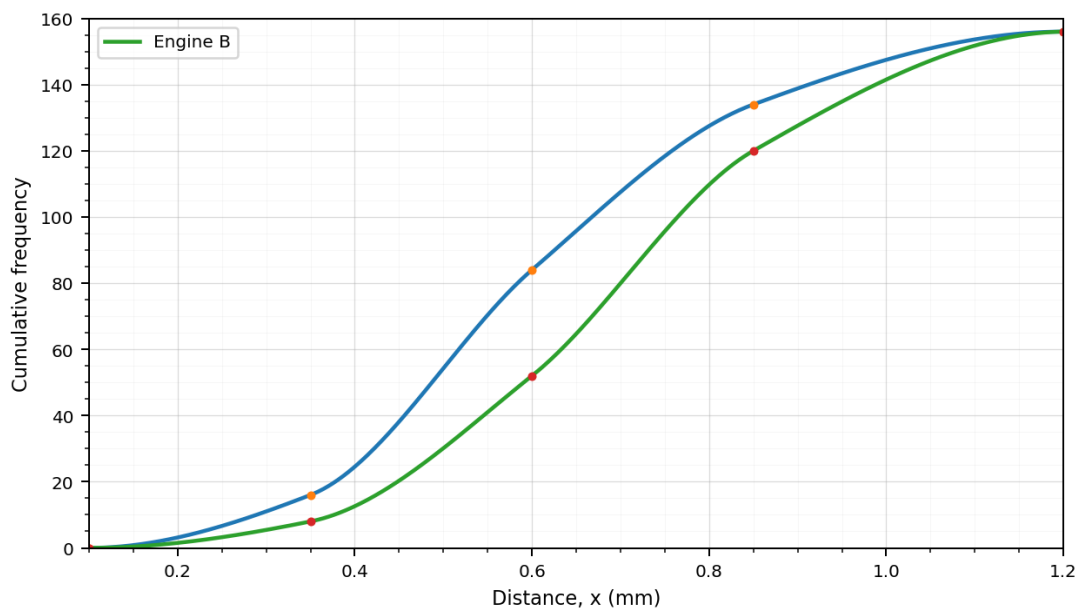
Width (cm)	< 9.5	< 14.5	< 19.5	< 29.5	< 39.5	< 44.5
No. books (cf)	0	3	16	41	65	70



(c)(i) At 27 cm, the graph gives a cumulative frequency of about 34 or 35 books.

(c)(ii) The widest 20 books start above the 50th value. The 50th value is about 33.25 cm; the largest boundary is 44.5 cm. So the widest 20 are approximately 33.25 to 44.5 cm.

C3 Comparing two engine types



(b)(i) Between 0.30 and 0.70 mm: Type A approximately $118 - 11 = 107$ engines; Type B approximately $91 - 4 = 87$ engines.

(b)(ii) Less than 0.55 mm: approximately 62 Type A + 46 Type B = 108 engines.

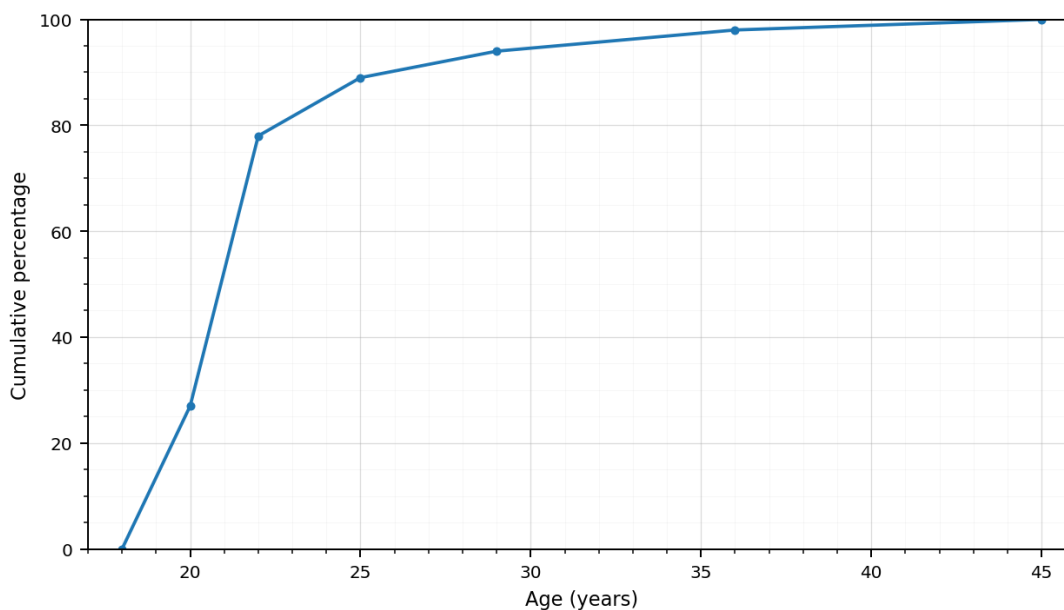
(c) If 16 Type A engines are above the repair limit, the fixed amount corresponds to about 0.81 mm. Reading Type B above the same value gives approximately 42 engines.

C4 Bank staff journey times

(a) Between 15 and 45 min: $76 - 11 = 65$ staff.

(b) 85% of 80 = 68, so $x = 30$ min. If 70% take y minutes or more, 30% = 24 staff take less than y ; hence $y = 20$ min. $(x + y)/2 = 25$ min. Number taking 25 min or more = $80 - 56 = 24$ staff.

C5 University ages and employment



(b) The oldest 8% means a cumulative percentage of 92%. The graph gives about 27.3-27.4 years, i.e. about 27 years 4-5 months.

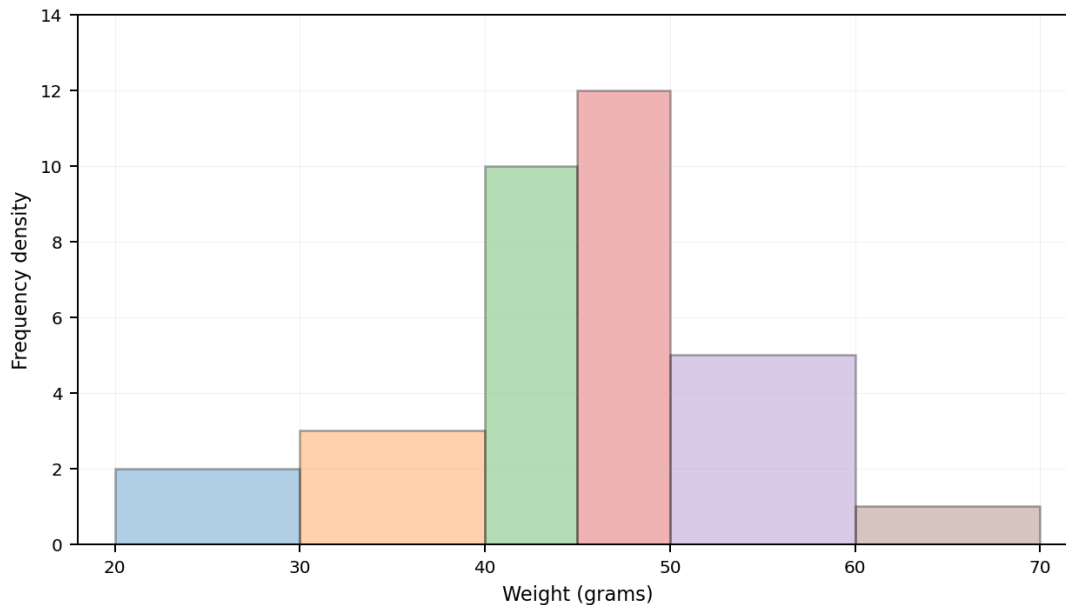
(c)(i) Among the 18-19-year-olds, $324 - 54 = 270$ are expected to find employment. This is $270/324 = 5/6$. Estimated number for the whole university = $(5/6) \times (324/0.27) = 1000$ students.

(c)(ii) Assume all age groups are equally likely to find employment. This is only an approximation. It could be an underestimate if older students have useful work experience; it could be an overestimate if employment conditions deteriorate before students finish their courses.

SECTION D - END-OF-CHAPTER REVIEW

D1 Sausage weights

(i) $210 - 160 = 50$ sausages.



(ii) Histogram frequencies are 20, 30, 50, 60, 50 and 10 for the successive classes 20-30, 30-40, 40-45, 45-50, 50-60 and 60-70. The corresponding frequency densities are 2, 3, 10, 12, 5 and 1 sausages per gram.

D2 Unknown leaves a and b

The sum is $22.5 + (a + b)/10$. Since $a - b = 7$, possible digit pairs are (7,0), (8,1), (9,2). These give totals 23.2, 23.4 and 23.6 respectively. Only 23.6 rounds to 24, so $a = 9$ and $b = 2$.

D3 Application download times

(a) $66 - 18 = 48$ downloads.

(b) For 5-6 min, frequency density is $48/1 = 48$. For 6-10 min, frequency is $80 - 66 = 14$, so density = $14/4 = 3.5$. Column heights are proportional to frequency density: $h/3.5 = 9.6/48$, giving $h = 0.7$ cm.

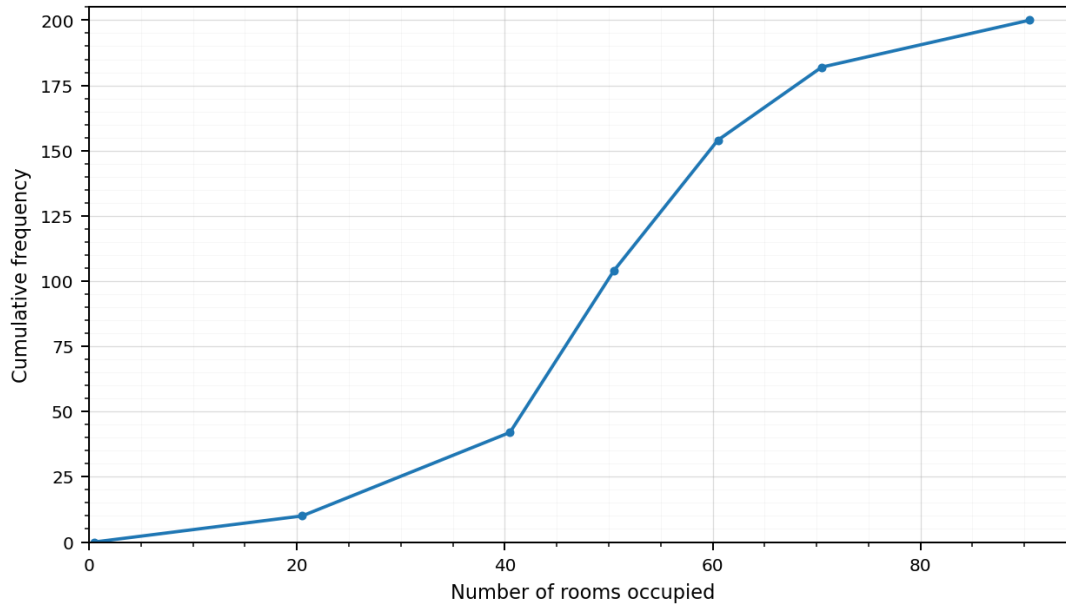
D4 Merging histogram classes

(a) Column areas are in the ratio $(1 \times 16) : (2 \times 12) : (4 \times 3) = 16 : 24 : 12 = 4 : 6 : 3$. Therefore the class frequencies are $(4/13)390 = 120$, $(6/13)390 = 180$ and $(3/13)390 = 90$.

(b) The two highest frequencies are the first two classes. Merged width = 3 and merged frequency = 300, so density = 100. The remaining density is $90/4 = 22.5$. If density 100 is drawn 30 cm high, remaining height = $(22.5/100) \times 30 = 6.75$ cm.

(c) The two lowest-frequency classes are the first and third classes, but the second class lies between them. They are not contiguous, so they cannot be merged into one continuous histogram class without also absorbing the middle class.

D5 Hotel room occupancy



(ii) Using a graph, the cumulative frequency at 30.5 rooms is about 20-26. Therefore the number of days with more than 30 rooms occupied is about $200 - cf = 174$ to 180 days.

(iii) 75% of $200 = 150$. At cumulative frequency 150, the graph gives about 58, 59 or 60 rooms.